

# Lesson 5: Solving Linear Inequalities

| Criteria                   | 4 - Exemplary                                                                | 3 - Proficient                                                            | 2 - Developing                                                     | 1 - Emerging                                                               |
|----------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------|
| Algebraic Solution Process | Solves for the variable using inverse operations, showing all steps clearly. | Solves for the variable with minor computational errors, but shows steps. | Attempts to solve but has consistent errors in inverse operations. | Cannot isolate the variable; little to no work shown.                      |
| Handling Negative Numbers  | Correctly reverses the inequality sign every time a negative is used.        | Reverses the sign most of the time but misses it once.                    | Identifies that a flip is needed but forgets to do it.             | Does not flip the inequality sign when multiplying/dividing by a negative. |
| Graphing on Number Line    | Correctly uses open/closed circles and shades in the correct direction.      | Uses correct circles but sometimes confuses shading direction.            | Confuses open/closed circles (e.g., uses closed for $<$ ).         | Graph does not correspond to the inequality.                               |

|                            |                                                                                            |                                                                                 |                                                                          |                                                                     |
|----------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------|
| Notation<br>(Interval/Set) | Accurately uses interval notation (e.g., $(-\infty, 2]$ ).                                 | Uses interval notation with minor errors (e.g., mixing brackets $[$ and $()$ ). | Uses set-builder notation incorrectly or makes multiple notation errors. | Unable to express the solution in any notation other than $x > a$ . |
| Checking<br>Solutions      | Verifies the answer by substituting values from the solution set back into the inequality. | Checks solution but does not write down the verification step.                  | Understands the need to check but makes errors during substitution.      | Does not check the                                                  |

## Lesson 1: Solving Quadratic Equations